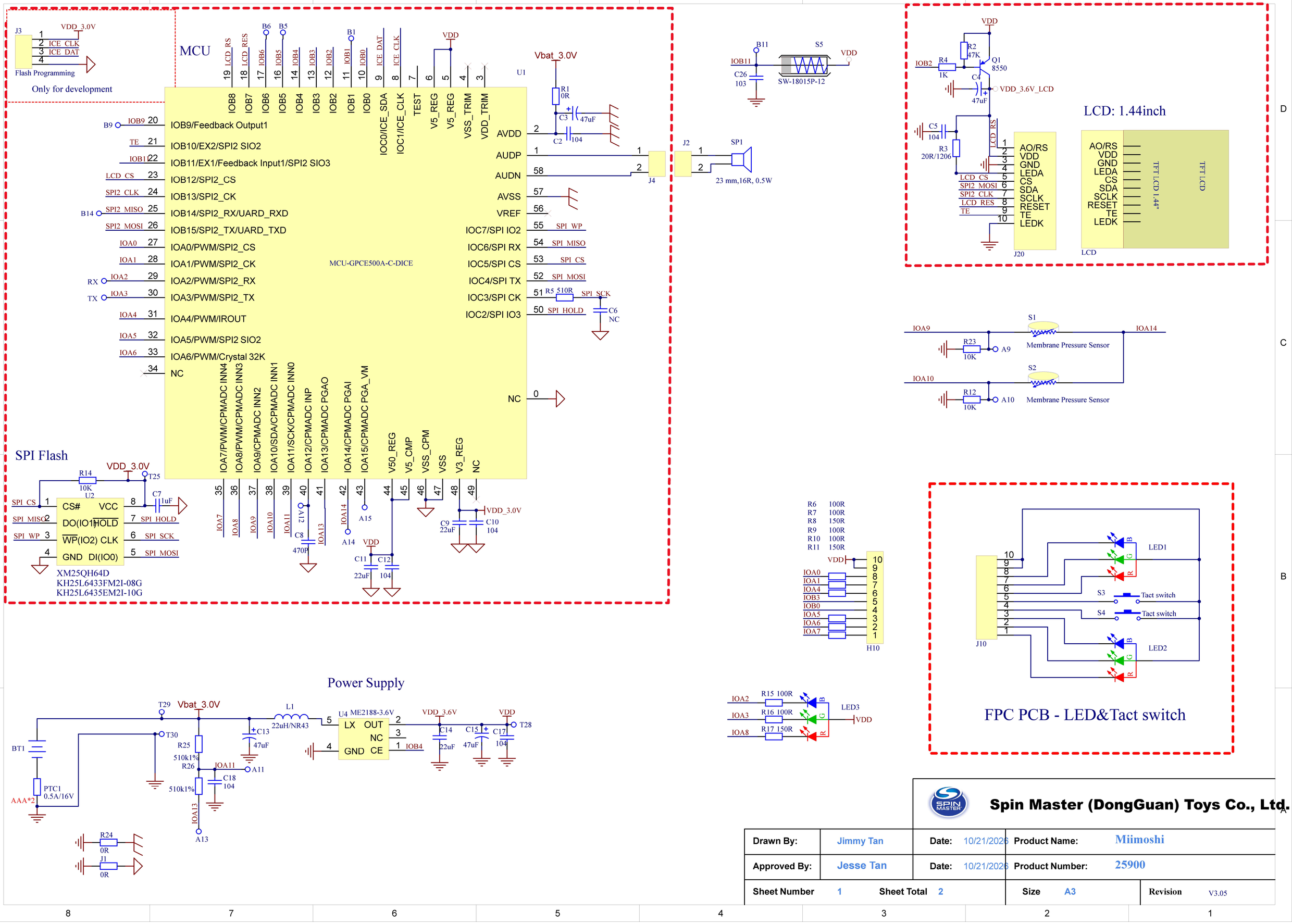

MCU
U1
MCU-GPCE500A-C-DICE
Only for development
Flash Programming
J3
VDD_3.0V
ICE_CLK
ICE_DAT
Vbat_3.0V
R1 0R
C3 47uF
C2 104
AVDD
AUDP
AUDN
AVSS
VREF
IOC7/SPI IO2
IOC6/SPI RX
IOC5/SPI CS
IOC4/SPI TX
IOC3/SPI CK
IOC2/SPI IO3
SPI_WP
SPI_MISO
SPI_CS
SPI_MOSI
SPI_SCK
R5 510R
C6 NC
J4
J2
SP1
23 mm,16R, 0.5W
B11
S5
IOB11
C26 103
SW-18015P-12
VDD
R2 47K
Q1 8550
R4 1K
IOB2
C4 47uF
VDD_3.6V_LCD
LCD: 1.44inch
C5 104
R3 20R/1206
LCD_RS
LCD_CS
SPI2_MOSI
SPI2_CLK
LCD_RES
TE
AO/RS
VDD
GND
LEDA
CS
SDA
SCLK
RESET
TE
LEDK
J20
TFT LCD 1.44"
TFT LCD
LCD
S1
S2
IOA9
IOA10
IOA14
R23 10K
R12 10K
A9
A10
Membrane Pressure Sensor
SPI Flash
XM25QH64D
KH25L6433FM2I-08G
KH25L6435EM2I-10G
R14 10K
C7 1uF
C8 470P
C11 22uF
C12 104
C9 22uF
C10 104
A12
A15
A14
T25
SPI_CS
SPI_MISO
SPI_WP
SPI_HOLD
SPI_SCK
SPI_MOSI
R6 100R
R7 100R
R8 150R
R9 100R
R10 100R
R11 150R
H10
J10
LED1
LED2
S3
S4
Tact switch
FPC PCB - LED&Tact switch
Power Supply
BT1
AAA*2
PTC1 0.5A/16V
T29
T30
T28
L1 22uH/NR43
U4 ME2188-3.6V
LX
OUT
NC
GND
CE
C13 47uF
C14 22uF
C15 47uF
C17 104
R25 510k1%
R26 510k1%
C18 104
A11
A13
R24 0R
J1 0R
R15 100R
R16 100R
R17 150R
LED3
Spin Master (DongGuan) Toys Co., Ltd.
Drawn By: Jimmy Tan
Date: 10/21/2026
Product Name: Miimoshi
Approved By: Jesse Tan
Date: 10/21/2026
Product Number: 25900
Sheet Number 1
Sheet Total 2
Size A3
Revision V3.05

Revision History

Revesion	Date	Reason	Description
V1.00	2026-06-26	Released initial scheamtic for quote and prototype validation.	1. Released complete schematic set (3 sheets) including main, LED&LCD and revision history sheet.
V1.01	2026-06-29	Project requirements change.	1. Changed R23&R24 to 3.3K 2. Connected LCD buttons with sockets 3. Marked accelerometer and its surroudings as NC 4. Added test pads for VBAT, IOA8, GND, IOB8, IOA13, IOA14. 5. Removed the leds which for LCD backlight. 6. Changed the LCD LED control IO from IOB0 to IOB1 (PWM IO) 7. Added two serial LEDs as an option 8. Two RGB LEDs are connected individually. 9. product names is updated to Momochi 10. Version is updated to V1.01
V2.00	2026-07-06	Project requirements change.	1.Main MCU was changed from GPCE6F to GPCE500A 2. Changed the LCD to JPY17160B-10P for QUOTE and LATF samples. 3. Added wake up pin for G-sensor 3. Kept the Booster output cap as low ESR MLCC for bettery ripple. 4. Kept two CAP closed to MCU as MLCC for quote at this stage due to space limitation.
V2.01	2026-7-7	Add socket to battery connection and change the project name	1. Added sockets (J3&J21) to battery connection instead of wire soldering. 2. Changed the project name to Miimoshi
V2.02	2026-7-8	Project requirements change.	1. Removed G-senor 2.Adeed Force Sensors on Analog pins(IOA8)
V2.03	2026-07-09	Project requirements change.	1. The serial LEDs were changed from 4pins to 6 pins; 2. Added C3 and removed C1
V2.04	2026-08-10	Project requirements change.	1.LCD changed to 1.44 inch ; 2. Speaker changed to 23mm16R0.5W ;
V2.05	20260902	Design update(Project requirements change).	1. LEDs and tach switch placed on sample PCB baord.
V3.01	2026/09/14		1. 2 Force Sensors be connected to PWM pins(IOA8) to supply power,
V3.02	2026/09/17		1. remove LDO, change LED resistor R6, R7, R9, R10 valuate to 100R. 2. continue using the Parallel LEDs
V3.04	2026/10/09		1. Add a third LED, 2. change switch&LED PCB to FPC PCB, add a FPC socket, 3. change pressure sensor size to 14mm*40mm L42mm
V3.05	2026/10/21		1. Relocate all the LED pins on PWM pins (A0 – A8), 2. change Pressure sensor power supply from IOA8 to IOA14

Spin Master (DongGuan) Toys Co., Ltd.

Drawn By:	JimmyTan	**Date:**	10/21/2026	**Product Name:**	Miimoshi		
Approved By:	Jesse Tan	**Date:**		**Product Number:**	25900		
Sheet Number	2	**Sheet Total**	2	**Size**	A3	**Revision**	V3.05